

CUSTOMER CHURN PREDICTION(SQL)

1. Churn Rate by Contract Type

Input:

```
Run SQL query/queries on table teleco_churn.cust_churn: ⓘ  
  
1 SELECT  
2     Contract,  
3     COUNT(*) AS total_customers,  
4     SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) AS churned,  
5     ROUND(  
6         100.0 * SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2  
7     ) AS churn_rate_pct  
8 FROM cust_churn  
9 GROUP BY Contract  
10 ORDER BY churn_rate_pct DESC;
```

Output:

Contract	total_customers	churned	churn_rate_pct ▾ 1
Month-to-month	7750	3310	42.71
One_year	2946	332	11.27
Two_year	3390	96	2.83
Contract	2	0	0.00

2. Customer Lifetime Value (CLV) Segmentation

Input:

Run SQL query/queries on table `teleco_churn.cust_churn`: ?

```
1 SELECT
2     CASE
3         WHEN Total_Charges >= 5000 THEN 'High Value'
4         WHEN Total_Charges BETWEEN 2000 AND 4999 THEN 'Mid Value'
5         ELSE 'Low Value'
6     END AS clv_segment,
7     COUNT(*) AS customer_count,
8     ROUND(AVG(Monthly_Charges), 2) AS avg_monthly_charges,
9     ROUND(AVG(Tenure_Months), 1) AS avg_tenure_months,
10    SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) AS churned_count
11 FROM cust_churn
12 GROUP BY clv_segment
13 ORDER BY avg_monthly_charges DESC;
```

Output:

clv_segment	customer_count	avg_monthly_charges	1	avg_tenure_months	churned_count
High Value	2270	99.03		65.2	314
Mid Value	3442	78.58		45.1	746
Low Value	8376	49.78		18.2	2678

3. Revenue at Risk from High-Value Churners

Input:

Run SQL query/queries on table `teleco_churn.cust_churn`:

```
1 SELECT
2     Contract,
3     COUNT(*) AS at_risk_customers,
4     ROUND(SUM(Monthly_Charges), 2) AS monthly_revenue_at_risk,
5     ROUND(AVG(Tenure_Months), 1) AS avg_tenure_months
6 FROM cust_churn
7 WHERE Churn_Label = 'Yes'
8     AND Monthly_Charges > (SELECT AVG(Monthly_Charges) FROM cust_churn)
9 GROUP BY Contract
10 ORDER BY monthly_revenue_at_risk DESC;
```

Output:

Contract	at_risk_customers	monthly_revenue_at_risk ▾ 1	avg_tenure_months
Month-to-month	2374	204483.20	16.2
One_year	260	25059.80	48.5
Two_year	76	7577.00	63.3

4. Top Cities by Churn (Geographic Risk Map)

Input:

Run SQL query/queries on table `teleco_churn.cust_churn`:

```
1 SELECT
2   City,
3   COUNT(*) AS total_customers,
4   SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) AS churned,
5   ROUND(
6     100.0 * SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2
7   ) AS churn_rate_pct,
8   ROUND(SUM(CASE WHEN Churn_Label = 'Yes' THEN Monthly_Charges ELSE 0 END), 2) AS lost_revenue
9 FROM cust_churn
10 GROUP BY City
11 HAVING COUNT(*) >= 50
12 ORDER BY churn_rate_pct DESC
13 LIMIT 15;
```

Output:

City	total_customers	churned	churn_rate_pct	1	lost_revenue
Modesto	56	24	42.86		1740.40
Santa_Barbara	56	20	35.71		1721.40
San_Diego	300	100	33.33		7057.80
Glendale	80	26	32.50		1267.10
Torrance	50	16	32.00		1308.20
Pasadena	60	18	30.00		1406.20
San_Francisco	208	62	29.81		4953.20
Los_Angeles	610	180	29.51		13241.40
San_Bernardino	56	16	28.57		1098.60
Berkeley	64	18	28.13		1456.80
Stockton	88	24	27.27		1892.70
San_Jose	224	58	25.89		4063.10
Oakland	104	26	25.00		1998.50
Riverside	64	16	25.00		1001.20
Fresno	128	32	25.00		2315.00

5. Impact of Security & Support Add-ons on Churn

Input:

Run SQL query/queries on table `teleco_churn.cust_churn`:

```
1 SELECT
2   Online_Security,
3   Tech_Support,
4   COUNT(*) AS total_customers,
5   ROUND(
6     100.0 * SUM(CASE WHEN Churn_Label = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2
7   ) AS churn_rate_pct,
8   ROUND(AVG(Monthly_Charges), 2) AS avg_monthly_charges
9 FROM cust_churn
10 WHERE Internet_Service != 'No'
11 GROUP BY Online_Security, Tech_Support
12 ORDER BY churn_rate_pct;
```

Output:

Online_Security	Tech_Support	total_customers	churn_rate_pct	1	avg_monthly_charges
Online_Security	Tech_Support	2	0.00		0.00
Yes	Yes	2198	9.01		81.47
Yes	No	1840	21.30		75.69
No	Yes	1890	22.33		79.76
No	No	5106	48.96		74.19